TCS A2A Catalog Reference

This document serves as the authoritative reference for Air-to-Air (A2A) combat elements within the Theater Control System (TCS). It details difficulty tiers, package compositions, and the aircraft roster used for dynamic spawning.

1. Bandit Tiers (Difficulty)

Tiers define the skill level of the AI pilots and the complexity of the weapons they are cleared to employ.

Tier	Label	AI Skill	Weapon Constraints	Description
A	**Beginner**	Average	Guns + Rear-aspect IR	Focuses on basic BFM and fundamental defensive reactions. No Fox-3.
G	**Standard**	Good	All-aspect IR + Fox-1	Introduces tactical intercepts and semi-active radar missile employment.
H	**Advanced**	High	Full Doctrinal (inc. Fox-3)	High-lethality environment with coordinated tactics and modern missiles.
X	**Expert**	Excellent	Maximum Threat (inc. Fox-3)	"Boss" level difficulty. Elite AI, aggressive timelines, and active ECM.

2. Package Definitions

Packages define the quantity and formation of the aircraft spawned within a single group.

Code	Size	Common Formations	Typical Roles
1S	Single Ship	Solo	BFM Training, Target Drones, 1v1 Intercepts
2S	Two-Ship	Line Abreast, Welded Wing	Standard tactical elements, Section leads
4S	Four-Ship	Tactical Spread, Wall, Box	Major intercepts, Offensive Counter Air (OCA)

3. Aircraft Catalog

These units are selected based on the Tier requested and the current `MISSION_YEAR_FLAG`.

ID / Type	Name	Primary Role	Era	Threat
MIG21	MiG-21bis	Interceptor	1959	LOW
MIG23	MiG-23MLA	Interceptor	1970	MED
MIG29	MiG-29 (9.12/9.13)	Multi-role	1983	MED

ID / Type	Name	Primary Role	Era	Threat
SU27	Su-27 Flanker-B	Air Superiority	1985	HIGH
SU33	Su-33 Flanker-D	Carrier-based Multi-role	1998	HIGH
J11	J-11A	Air Superiority	1998	HIGH
M2000	Mirage 2000C	Interceptor	1984	MED
F5	F-5E Tiger II	Light Fighter	1972	LOW
F14	F-14B/A Tomcat	Heavy Interceptor	1974	HIGH
JF17	JF-17 Thunder	Multi-role	2007	HIGH
MIG31	MiG-31 Foxhound	Heavy Interceptor	1981	EXTREME
TU22M	Tu-22M3 Backfire	Bomber	1972	LOW (Defensive)
TU95	Tu-95 Bear	Strategic Bomber	1956	LOW (Defensive)

4. Global Loadout & Engagement Rules

The system enforces weapon restrictions based on the Tier to ensure training objectives are met.

Weapon Access by Tier

Weapon Type	Beginner (A)	Intermediate (G)	Advanced (H)	Boss (X)
Internal Gun	☑ Yes	☑ Yes	☑ Yes	☑ Yes
Rear-Aspect IR	☑ Yes	☑ Yes	☑ Yes	☑ Yes
All-Aspect IR	❌ No	☑ Yes	☑ Yes	☑ Yes
Fox-1 (SARH)	❌ No	☑ Yes	☑ Yes	☑ Yes
Fox-3 (ARH)	❌ No	❌ No	☑ Yes	☑ Yes
ECM / Jamming	❌ No	❌ No	☑ Yes	☑ Yes

Intent-Based Altitudes

Intent	Altitude Band	Typical MSL
WVR / BFM	LOW	15,000 – 20,000 ft
BVR / CAP	MEDIUM/HIGH	25,000 – 35,000 ft
Interceptor	HIGH	35,000 – 45,000 ft
Bomber	MEDIUM	25,000 – 30,000 ft

5. Escort Package Details

Friendly packages used for escort missions follow a specific naming convention to allow script discovery.

Group Name	Aircraft Type	Speed (KTAS)	Role
PACKAGE_TANKER	IL-78 / KC-135	280–320	High Value Airborne Asset (HVAA)
PACKAGE_TRANSPORT	An-26 / Il-76	260–300	Logistics / Airborne Insertion
PACKAGE_STRIKE	Su-24 / Su-34	450–500	Precision Strike / Deep Interdiction

6. Machine-Readable Naming Convention

All A2A templates must follow the data contract for the injector to process them correctly:

`BANDIT_<ROLE>_<TIER>_<TYPE>_<PKG>[_<VAR>]`

Example: `BANDIT_BVR_H_SU27_4S`

- **Role**: BVR
- **Tier**: Advanced (H)
- **Type**: Su-27
- **Pkg**: 4-Ship

7. Operational Distances

Distances are calculated from the Session Lead to ensure realistic reaction timelines.

Mission Intent	Spawn Distance (NM)	Bearing Arc
Build Skills	5 – 15	Inline or Offset
ACM H2H	5 – 8	Head-on
BFM Fox-2	10 – 15	Head-on
CAP Defense	20 – 40	±90°
Intercept	40 – 80	±90° to ±135°
Sweep	60 – 100	±135°

This document is generated based on the TCS A2A Bandit Template Reference. Ensure any modifications to naming conventions are reflected in both documents.